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Dear Editors,

Please consider the transferred manuscript **Fragile Authentication of PDF Signatures via Bit-Exact Quantization Index Modulation on Vector Geometry** (transferred from **Signal Processing**, manuscript SIGPRO-D-26-04464) for publication in **Digital Signal Processing**.

Why this journal. The paper is a watermarking/signal-processing study, not a cryptography protocol paper: it asks how a fragile payload can be carried by the geometry of a visible mark, how exact (bit-for-bit) recovery can be achieved with dithered quantization index modulation, and what capacity–distortion trade-off results on a fixed 27-degree-of-freedom carrier. Those are core DSP questions, and the manuscript is now written for that readership.

What changed on transfer. The transferred file was the earlier submission. Before completing this submission I have revised it substantially: a new title and abstract framed for the DSP/watermarking readership; a rebuilt, verified bibliography (every citation resolves to a DOI, standard or book — the earlier draft contained references that could not be verified, and they were removed); a new PDF-level evaluation on a 180-file corpus from three document generators (LibreOffice, Typst, Chromium print) in native, flattened and print-scan-simulated variants, with a carrier ablation and same-carrier baselines; and the optical/screen-photo and API material moved out of this paper into a companion manuscript.

What the paper claims and does not claim. It reports exact recovery of a 266-bit payload, fragility to coordinate edits and re-serialisation, fail-closed behaviour under flattening, and a capacity–distortion comparison in which capacity-limited baselines are reported as such. It does not claim cryptographic security from a watermarking argument, and it explicitly complements rather than replaces PAdES/eIDAS signatures.

Declarations. The manuscript is not under consideration elsewhere; the preprint is archived on Zenodo (concept DOI 10.5281/zenodo.23429017) and SSRN (10.2139/ssrn.7631458); a Ukrainian patent application (a 2026 06266) covers the sealing method. The evaluation harness and corpus manifest are available for reproducibility.

Thank you for considering the transfer.

Sincerely,
Yaroslav V. Baienko